

22nd cycle - Spring School 2026
Design Methods and Processes
Introduction
14 March 2026

Gaetano Cascini



OUTLINE

Design Methods and Processes

- Context and scope
- Design and Science

Course Description

- Lectures Topics, Tutorials, Team Working
- Course Calendar & Organisation
- Assignment and evaluation

DESIGN METHODS AND PROCESSES: SCOPE

What is design?

- *What is the last thing you designed?*
- *What does it mean designing?*
- *What makes someone a designer?*



DESIGN METHODS AND PROCESSES: SCOPE

**Everyone can – and does – design
(N. Cross)**



- **We all design when we plan for something new to happen,** whether that might be a new version of a recipe, a new arrangement of the living room furniture, or a new layout of a personal web page
- **So, design thinking is something inherent within human cognition;** it is a key part of what makes us human.

DESIGN METHODS AND PROCESSES: SCOPE

Design is about effecting change

*“Engineers are not the only professional designers. Everyone designs who devises courses of action aimed at **changing existing situations into preferred ones**”* [Herbert Simon]

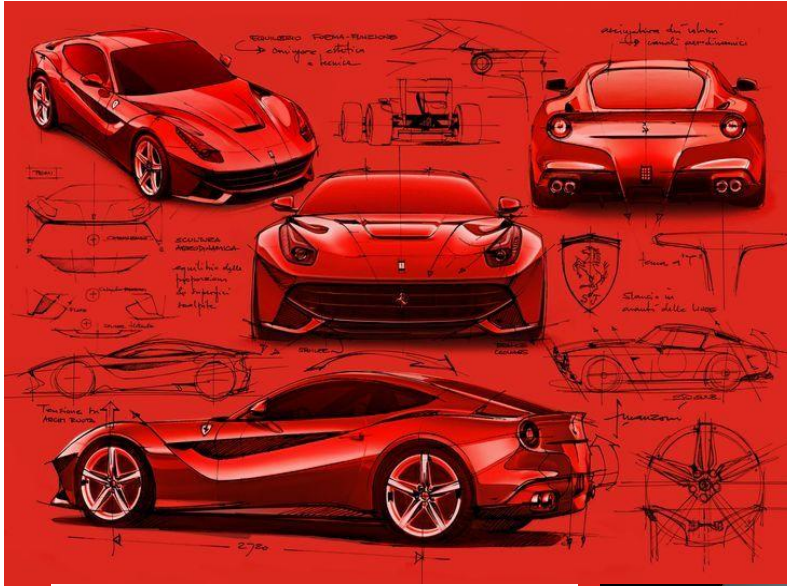
Design is about understanding how things work:

*“Design is a funny word. Some people think design means how it looks. But of course, if you dig deeper, it’s really **how it works**. To design something really well, you have to get it. You have to really grok [understand] what it’s all about.”* [Steve Jobs]

Design is about synthesis and creation

*“A designer is an emerging **synthesis** of artist, inventor, mechanic, objective economist and evolutionary strategist”* [R Buckminster Fuller]

DESIGN METHODS AND PROCESSES: SCOPE



What do these **artefacts** have in common?

DESIGN METHODS AND PROCESSES: SCOPE

Design Context: Human beings and Needs





DESIGN METHODS AND PROCESSES: SCOPE

Design Context: Human beings and Needs

Identifying needs is not just a matter of asking people

“If I had asked people what they wanted, they would have said faster horses” [attributed to Henry Ford]

Designers envision opportunities

“People don't know what they want until you show it to them. That's why I never rely on market research. Our task is to read things that are not yet on the page” [Steve Jobs]

New sources of inspiration emerge

“Information is the oil of the 21st century, and analytics is the combustion engine” [Peter Sondergaard]

DESIGN METHODS AND PROCESSES: SCOPE

Design Context: For Whom?

The multi stakeholder context of beyond use situations



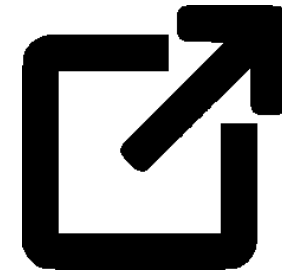
BUYER



USER



BENEFICIARY



OUTSIDER



Source: Cantamessa, Cascini, Montagna (2012)

DESIGN METHODS AND PROCESSES: SCOPE

Design Context: For Whom?

The multi stakeholder context of beyond use situations



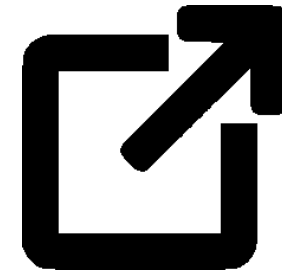
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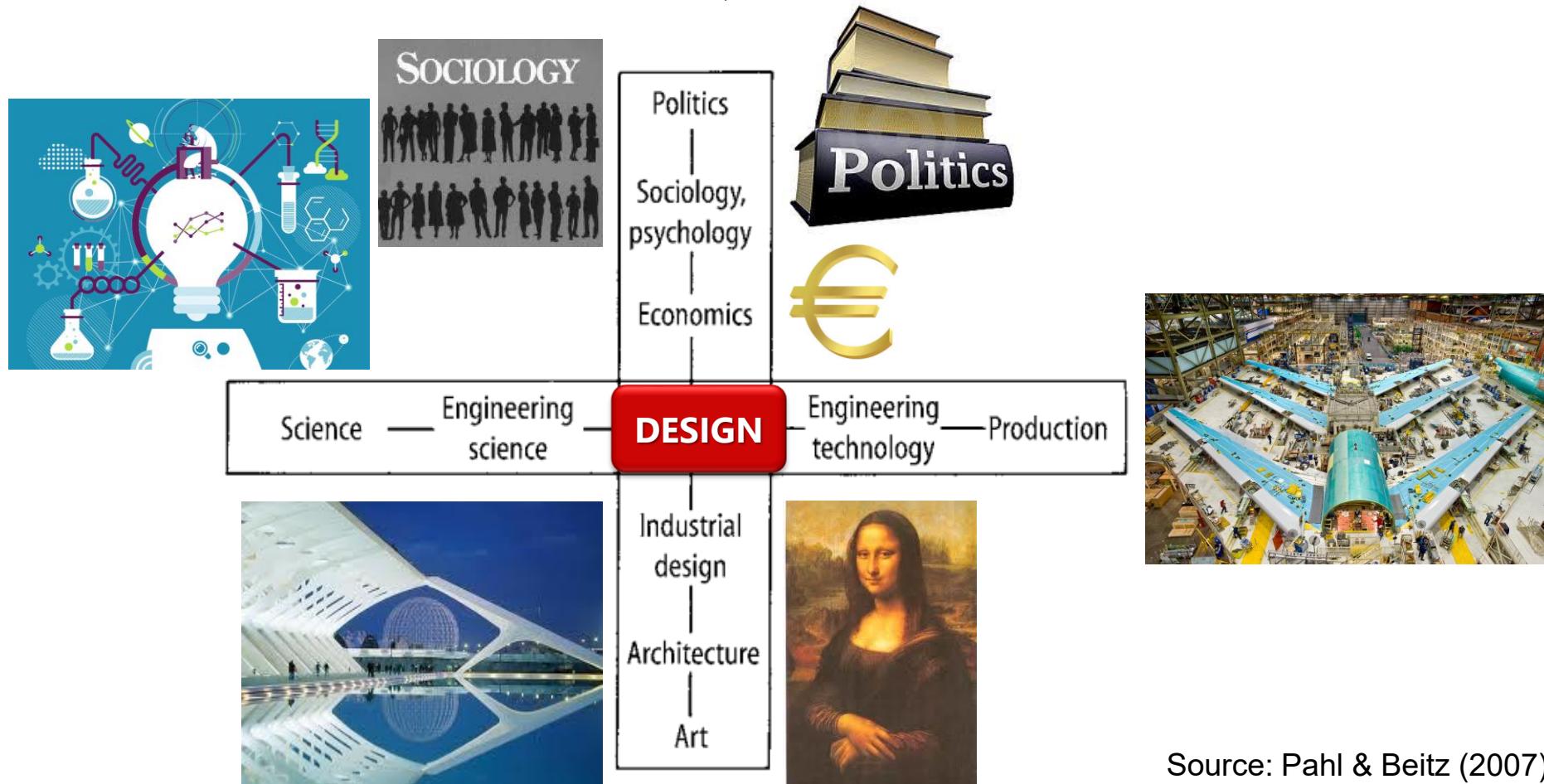
OUTSIDER



Source: Cantamessa, Cascini, Montagna (2012)

DESIGN METHODS AND PROCESSES: SCOPE

Design Context: From Art to Politics, from Science to Products



DESIGN METHODS AND PROCESSES: SCOPE



CAVE RELICS Clues to relatively modern behavior 44,000 years ago in Africa. Credit: Francesco d'Errico and Lucinda Backwell

Design has always relied on previous experience

DESIGN METHODS AND PROCESSES: SCOPE



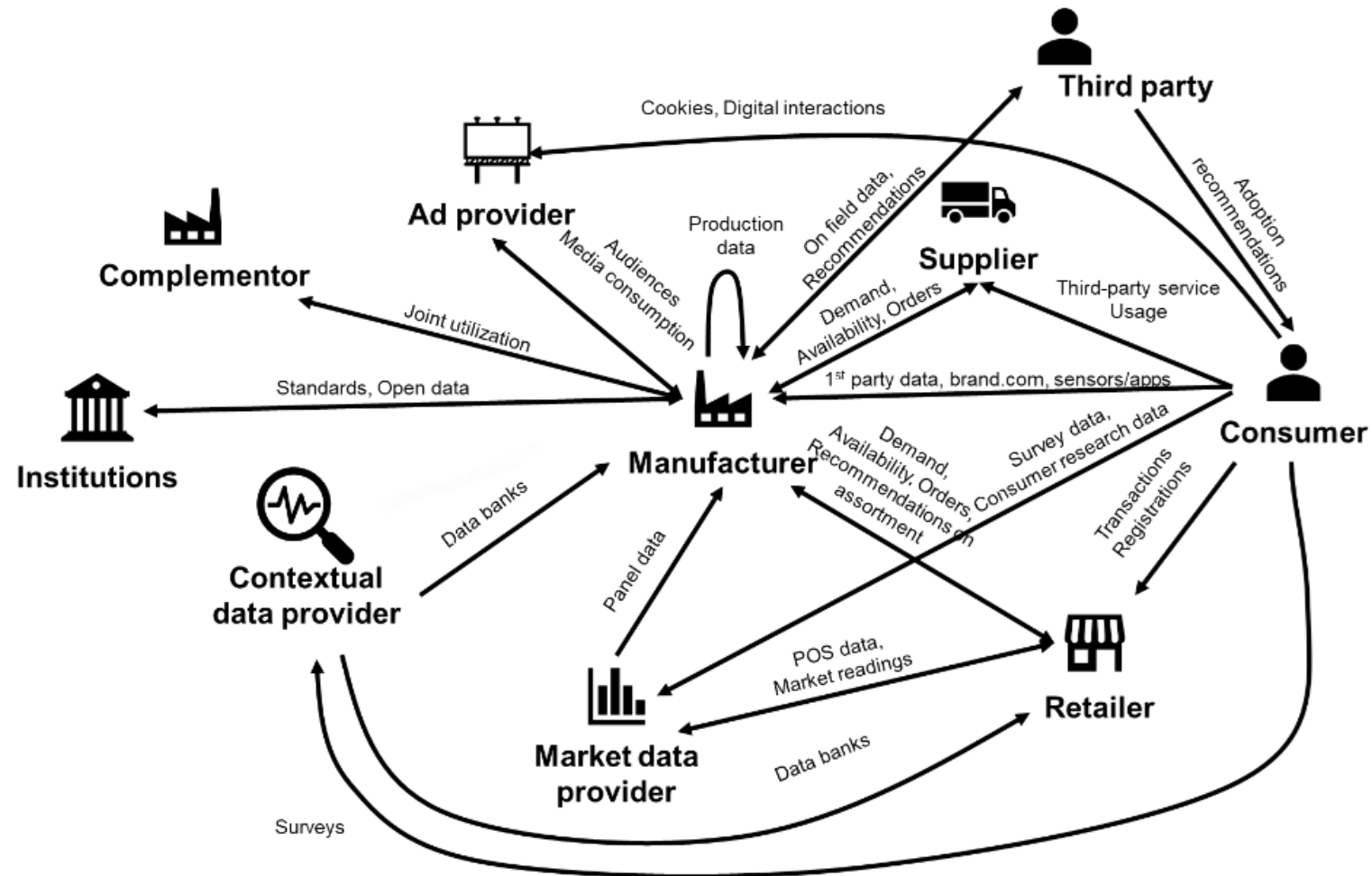
Design Context: Production

DESIGN METHODS AND PROCESSES: SCOPE



Design Context: Delivery

DESIGN METHODS AND PROCESSES: SCOPE



Design Context: Data flow network

DESIGN METHODS AND PROCESSES: SCOPE



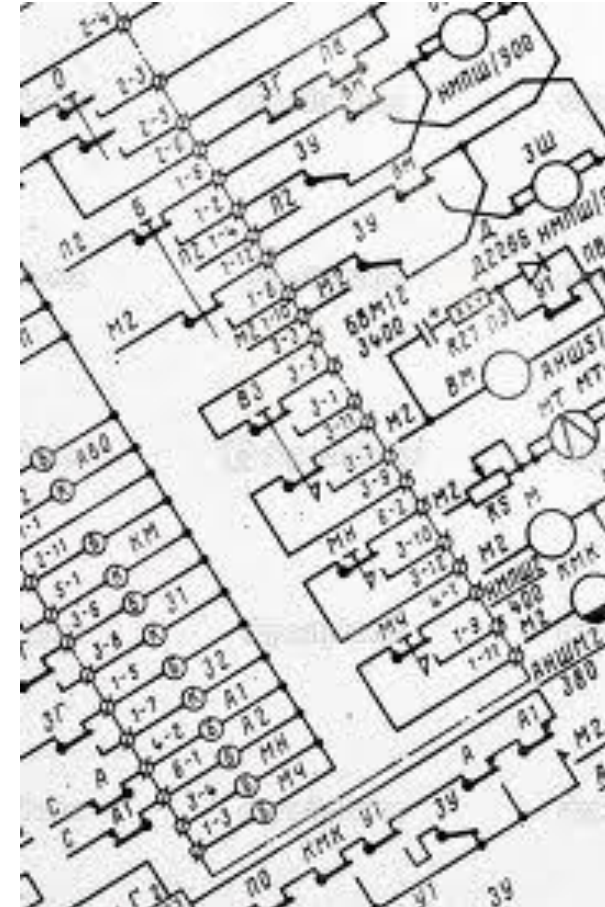
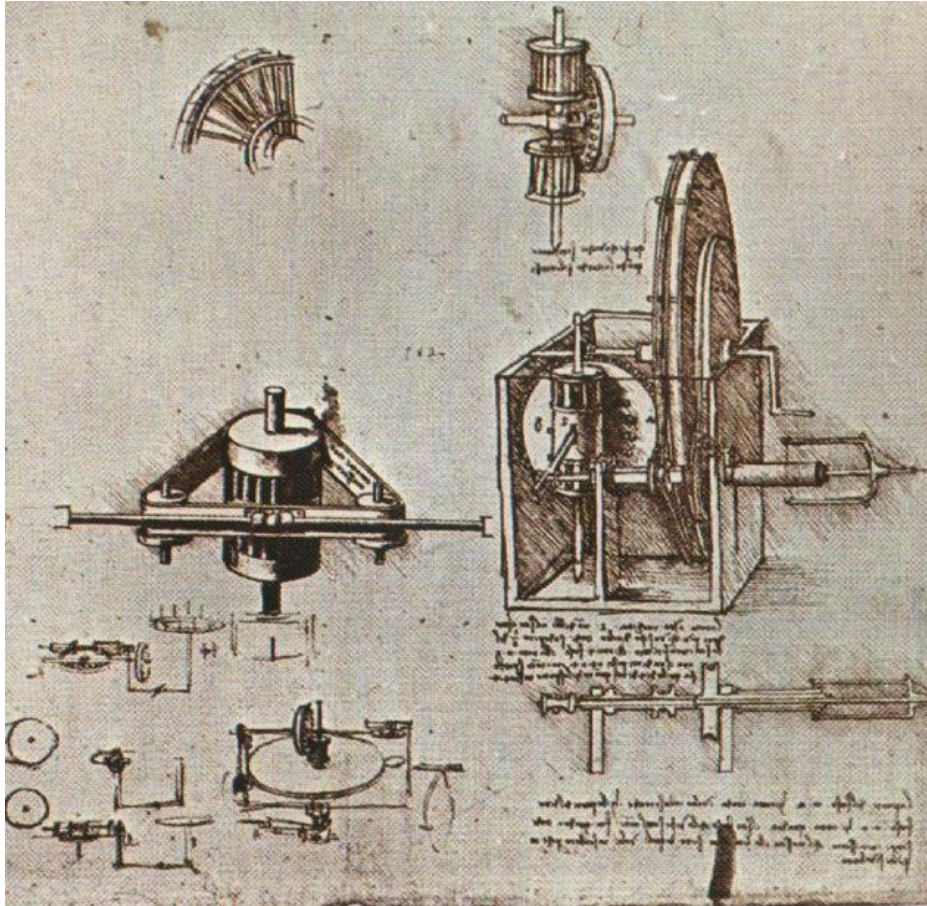
What are these **people** doing? What do they **share**?

DESIGN METHODS AND PROCESSES: SCOPE



How do people **share design information?**

DESIGN METHODS AND PROCESSES: SCOPE



Design Representations: Language for Technical Communication



DESIGN METHODS AND PROCESSES: SCOPE

The central role of design (C. McMahon)

Design is about synthesis, about creation with a holistic view

- Design is integrative – it brings together many viewpoints
- Design fosters understanding of constraints and trade-offs
- Design tells us where scientific effort is needed
- Design guides management
- Design can and should have a central role in guiding our future actions

Our lives are dominated by particular technologies

- Transportation, energy, buildings, manufacturing, healthcare...
- Design can guide their evolution and adaptation



DESIGN METHODS AND PROCESSES: SCOPE

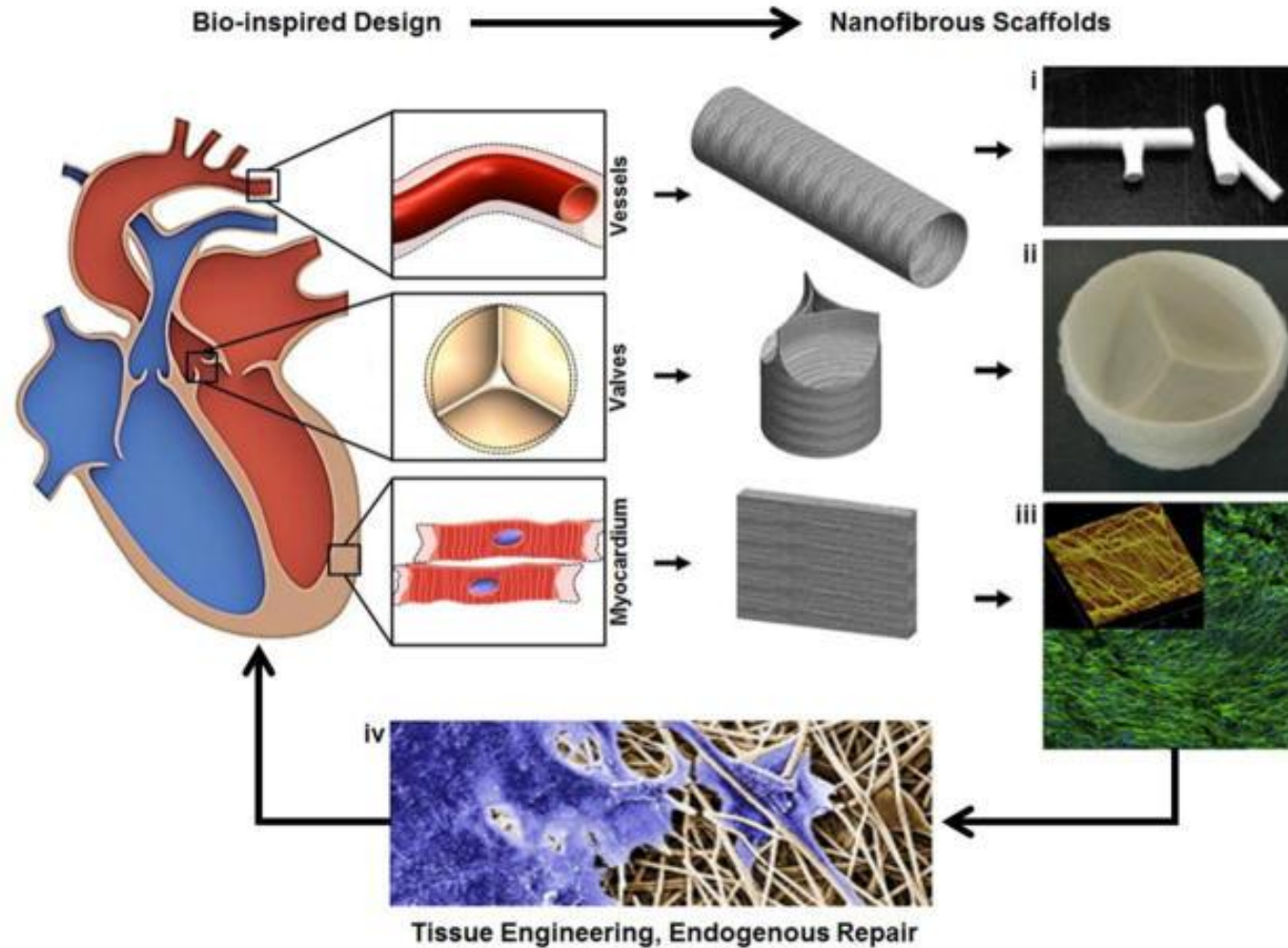
Relationships between **science** and **design** (Cross, 2001)

SCIENTIFIC DESIGN refers to modern, industrialized design and employs scientific knowledge resulting from natural and social sciences - both fundamental and applied - to engineering and architecture;

DESIGN SCIENCE has the aim of developing scientifically-based, domain-independent, explicitly organized, rational and systematic methods and tools to improve the design action;

SCIENCE OF DESIGN views the design process as a phenomenon to be studied scientifically, i.e. with systematic and reliable methods of investigation, assuming academic perspectives that can range from psychology to ethnography and from cognitive to organizational science.

SCIENTIFIC DESIGN (example)

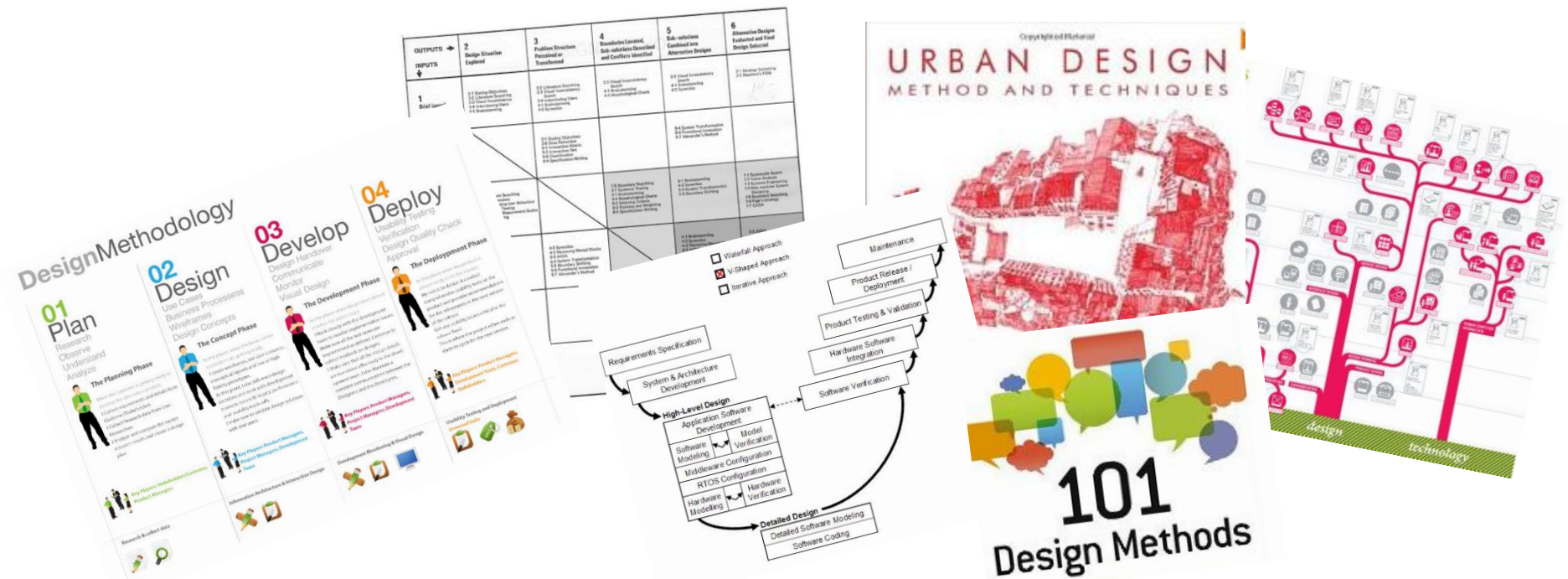


DESIGN SCIENCE (example)

Classification schemes

(John Clarkson)

- ❖ Stage-based vs Activity-based Models
- ❖ Solution-oriented vs Problem-oriented
- ❖ Abstract vs Procedural vs Analytical approaches



DESIGN SCIENCE (example)

ARTIFICIAL
INTELLIGENCE
X DESIGN

sig

COGNITIVE
DESIGN
SCIENCE

sig

COLLABORATIVE
DESIGN

sig

DATA
INFORMED
DESIGN

sig

DESIGN
CREATIVITY

sig

DESIGN
EDUCATION

sig

DESIGN FOR
ADDITIVE
MANUFACTURING

sig

DESIGN
JUSTICE
& ETHICS

sig

DESIGN
PRACTICE

sig

DESIGN
PROCESS

sig

DESIGN
RESEARCH
QUALITY

sig

DESIGN
SKETCHING

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DESIGN
THEORY

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HEALTH
SYSTEMS
DESIGN

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HUMAN
BEHAVIOUR
IN DESIGN

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PRODUCT
SERVICE
SYSTEMS

sig

ROBUST
DESIGN

sig

SUSTAINABLE
DESIGN

sig



SCIENCE OF DESIGN (example)



Design ability: natural born gift or something to learn?

- ❖ **Experienced** designers vs. **novice** designers
- ❖ Different **backgrounds** thus different design **behaviours**?
- ❖ Inherent **human** cognitive activity vs. **Artificial** Intelligence

DESIGN METHODS AND PROCESSES



Lectures
& Talks



Tutorials



Team
Working

DESIGN METHODS AND PROCESSES PROGRAMME

March 14th, 9:30-16:15 Politecnico di Torino, Castello del Valentino, Classroom 1V

9:30 Engineering Design, what are we talking about?

Gaetano Cascini, Politecnico di Milano

10:45 Morning Break

11:00 Design for Technology Adoption

Deborah Nas, TU Delft/deborahnas.com

12:00 From Data & AI to Value Creation - Designing products, services, and human-machine collaboration

Andrea De Mauro, Executive Advisor

13:00 Lunch Break

14:00 Tutorial: Stakeholders identification

Gaetano Cascini and Tutors

14:45 Practice session: Stakeholders identification

Tutors

16:15 Closing session

Gaetano Cascini, Politecnico di Milano



DESIGN METHODS AND PROCESSES PROGRAMME

April 18th, 9:30-16:15 Politecnico di Milano, Via Bonardi - Edificio 16B, Classroom 16B.1.1

9:30 *Innovating today – why design theory matters*

Pascal Le Masson, MINES ParisTech – PSL Research University



10:45 Morning Break

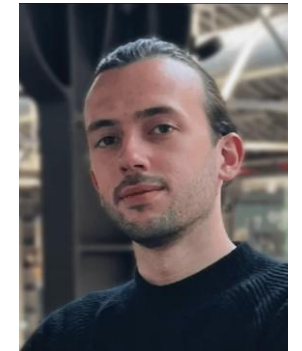
11:00 *Design Theory workshop*

Pascal Le Masson, Chipten Valibhay, MINES ParisTech – PSL Research University

12:45 Lunch Break

13:45 LLMs-Enhanced Creativity in Design: Potential, Risks, and Responsible Use

Filippo Chiarello, University of Pisa



15:00 *Pivot*

Lapo Peruzzi, Postura Ergonomics

15:45 *Closing remarks and preparation for the residential school*

Gaetano Cascini, Politecnico di Milano

DESIGN METHODS AND PROCESSES PROGRAMME

Oropa - 4-6 May 2026

- Hands On Problem Framing and Idea Generation
- Hands On Concept Formulation and Assessment
- Team working on your ASP Project



DESIGN METHODS AND PROCESSES: EVALUATION

Students' Essays (Deadline XXX September 2025)

- ❖ Discuss your design experience against an assigned topic of the spring school

Assessment Criteria:

- Scope: relevance to the proposed theme
- Presentation and style: adequate structure, clarity, length
- Content: completeness, originality, correctness

- Bonus-Malus: teamwork activity in May

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Thanks for your time!

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